3. DESIGN APPROACH

Aerial Matrix™ is designed to enhance search and rescue and emergency response operations by providing a coordinated, autonomous aerial lighting and visual assistance system. The system utilizes a matrix of drones that operate collaboratively to improve visibility, situational awareness, and operational efficiency in low-light or no-light environments. By integrating wireless communication, object detection, and modular payload systems, Aerial Matrix™ enables hands-free illumination and adaptive support for field personnel.

Throughout the design process, multiple system requirements, constraints, and standards guided development decisions. The most critical requirements include the ability to detect and track multiple objects, maintain synchronized autonomous flight across multiple drones, and support modular attachments that can be rapidly deployed. Key constraints include system cost, weight limitations, power consumption, and environmental durability. Additionally, compliance with standards such as IEEE 802.11 for wireless communication and FAA regulations for unmanned aerial systems influenced the design. The following subsections describe the evaluated design approaches and justify the selected system architecture.

3.1 Design Options

For the design approach, the team evaluated two architectural solutions for implementing the Aerial Matrix™ system. These approaches differ in how computational responsibilities are distributed between onboard components, particularly for flight control, communication, and object detection.

The first approach utilizes a fully distributed architecture in which each drone relies solely on a microcontroller to perform all required tasks, including flight stabilization, communication, and limited onboard processing. This design prioritizes low power consumption, reduced system weight, and cost efficiency, making it highly scalable for multi-drone deployment.

The second approach introduces a heterogeneous architecture that separates control and perception tasks. In this configuration, a microcontroller is responsible for real-time flight control and communication, while a dedicated AI accelerator performs computationally intensive object detection and target tracking. This separation improves system capability and enables real-time perception, but introduces additional integration complexity, cost, and power considerations.

These two approaches represent a trade-off between system efficiency and computational performance, which guided the selection of the final system architecture.

3.1.1 Design Option 1

One design option considered was a fully distributed system based solely on a microcontroller. In this configuration, each drone would operate using a microcontroller to handle flight control, wireless communication, and all onboard processing tasks. The ideal microcontroller must include integrated Wi-Fi capabilities, allowing drones to communicate directly without requiring additional networking hardware. Microcontrollers are compact, power-efficient, and cost-effective. These qualities make it well-suited for multi-drone deployment where weight and energy efficiency are critical.

A major benefit of this design is its simplicity and scalability. Because each drone operates independently, additional drones can be added to the system without significant architectural changes. Microcontrollers are generally low cost, which allows for a larger number of drones to be deployed within the project's budget constraints. Furthermore, the reduced power requirements support longer flight times, which is essential for search and rescue operations.

However, this approach presents significant limitations in computational capability. Current microcontrollers are not designed to handle advanced image processing or real-time object detection at the level required for autonomous tracking. Implementing these features solely on the microcontroller would result in reduced accuracy, slower response times, or the inability to meet system requirements. Because object detection is a core requirement of Aerial Matrix™, this limitation led to the consideration of an alternative design.

3.1.2 Design Option 2

The selected design approach incorporates a hybrid system that combines the microcontroller with a dedicated AI processing unit in the form of an attachment. In this configuration, the microcontroller is responsible for drone control, communication, and subsystem management, while an AI accelerator performs object detection and target tracking.

This design provides significant advantages over the fully microcontroller-based approach. The chosen AI accelerator optimizes machine learning inference, allowing for efficient and accurate real-time object detection. By offloading computationally intensive vision tasks to this AI accelerator, the system can meet the requirement of tracking multiple objects while maintaining fast response times. Additionally, the AI accelerator that we will choose should be conservative with power and small in size to help alleviate weight constraints.

Another advantage of this approach is the separation of responsibilities between system components. The microcontroller maintains reliable flight control and communication, while the AI accelerator handles perception. This division improves system stability and allows each component to operate within its optimal performance range. Furthermore, this architecture supports future scalability, as improvements to object detection algorithms can be implemented without modifying the control system.

Despite these advantages, this design introduces additional complexity due to the integration of multiple processing units. It also increases system cost compared to a purely microcontroller-based solution. However, these trade-offs are justified by the significant improvement in system capability and the ability to meet the core functional requirements of autonomous tracking and target identification while staying modular and scalable in design.

3.2 System Overview

Figure 3-1 presents a Level 0 system overview of the Aerial Matrix™ platform. At this level, the system is treated as a single functional unit, showing the primary inputs and outputs without internal detail. User input and environmental data, including target presence and ambient conditions, are received by the system. Internally, the system processes this information using onboard control, sensing, and communication subsystems powered by an integrated battery. The system outputs include positional data and visual feedback delivered to the mobile application interface. The system encapsulates all onboard components including power supply, processing units, sensing modules, and communication interfaces, defining a clear boundary between internal functionality and external user interaction.

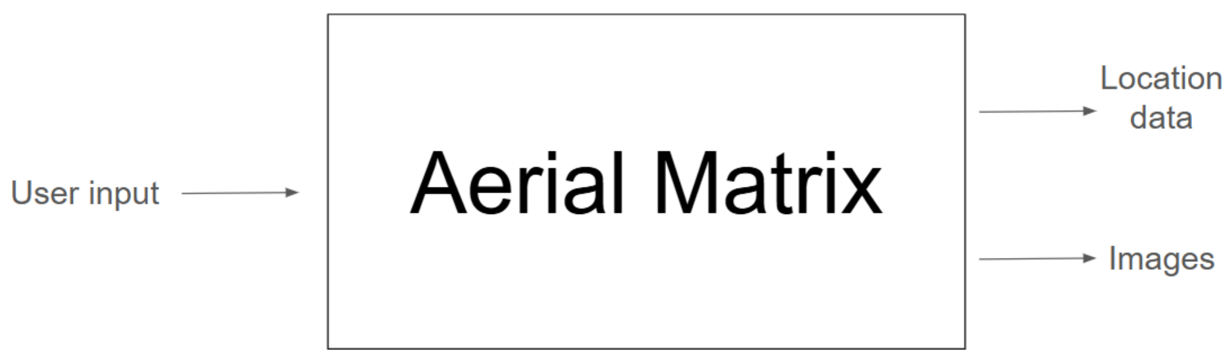

Fig. 3-1: Aerial Matrix™ at a Glance (Level 0)

This high-level representation highlights the overall system boundaries and primary functional relationships. It establishes a clear understanding of how external inputs are transformed into actionable outputs, providing a foundation for the more detailed subsystem interactions shown in subsequent diagrams.

Figure 3-2 expands upon the Level 0 overview by illustrating the internal subsystems and their interactions. This Level 1 diagram identifies key components including the microcontroller, AI camera, sensors, motor control system, and power regulation modules, along with the flow of data and power between them.

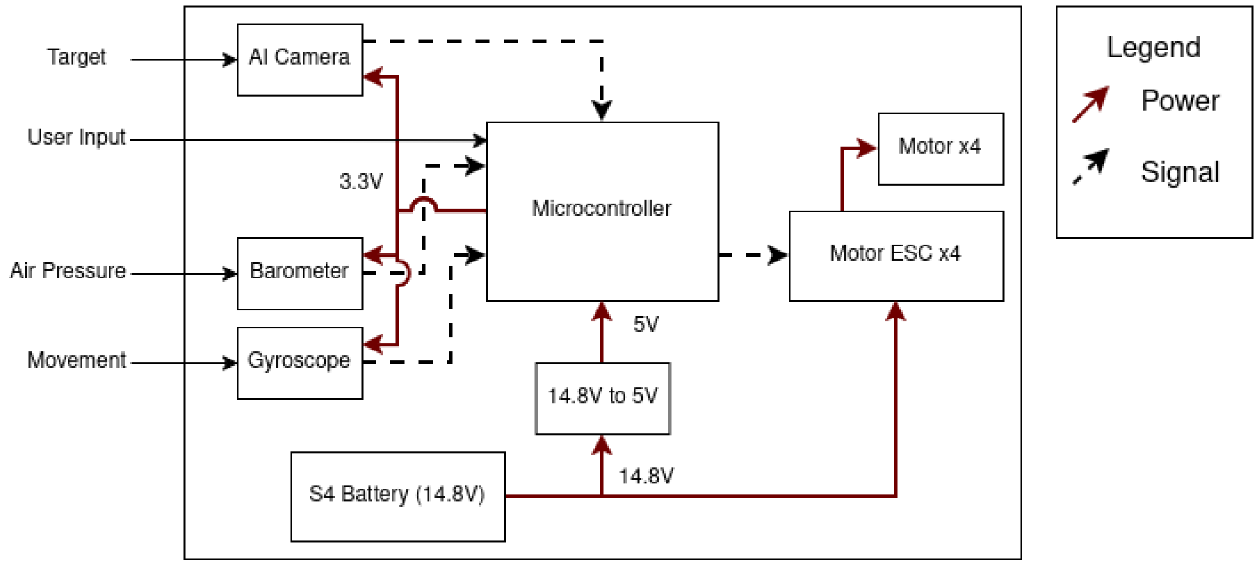

Fig. 3-2: Aerial Matrix™ Functionality (Level 1)

The Level 1 diagram demonstrates how the Aerial Matrix™ system integrates sensing, processing, and actuation components. The microcontroller serves as the central controller, receiving inputs from sensors and the AI camera, processing this data, and issuing control signals to the motor system. Power distribution from the battery through voltage regulation ensures stable operation across all components. This structure enables coordinated flight control and real-time response to user input and environmental conditions.

3.2.1 Microcontroller

For the Aerial Matrix™ system, several different microcontroller and embedded processing platforms were evaluated based on their computational capability, power consumption, wireless communication features, and cost. Because the system requires multiple drones operating simultaneously, each unit must be lightweight, energy efficient, and capable of reliable wireless communication. Table 3-1 shows the specifications of the processing platforms considered.

Table 3-1: Microcontroller Options

Processor	Clock speed (MHz / GHz)	Wi-Fi Capability	Power Consumption (watts)	Price (USD)	Suitability
Requirements	≥ 160 MHz (estimated)	Built-in Preferred	≤ 2 W (target)	< $100	Lightweight, Realtime control
ESP-S3 Dev Board [1]	240 MHz	Yes	1	$10 - $15	Excellent
Arduino Uno [2]	16 MHz	No	2.5	$20	Poor
Arduino Uno Q [3]	48 MHz	Yes	1.9	$22	Great
Raspberry Pi4 Model B 4GB [4]	1.5 GHz	Yes	3 – 7	$60 - $110	Overqualified

Orion Jetson Nano Super [5]	1.43 GHz	Yes	5 – 10	$150 - $200	Overkill

The listed design targets represent estimated performance goals based on system timing and operational constraints. These values are subject to refinement through further analysis and testing, particularly with respect to control loop timing and processing requirements.

The ESP32-S3 development board was selected as the primary microcontroller for each drone in the Aerial Matrix™ system due to its balance of performance, power efficiency, and integrated wireless capability.

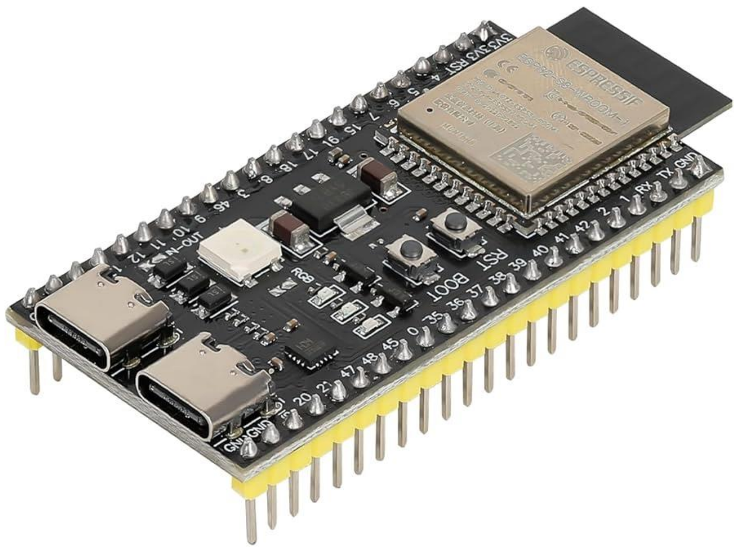

Fig. 3-3: ESP-S3 Dev Board

With a dual-core processor operating at up to 240 MHz and built-in Wi-Fi support, the ESP32-S3 is capable of handling real-time flight control, inter-drone communication, and subsystem coordination without requiring additional networking hardware.

Compared to Arduino-based platforms, the ESP32-S3 offers significantly higher processing capability and native wireless communication, eliminating the need for external modules and reducing system complexity. While Arduino boards provide low power consumption, their limited clock speed and lack of integrated networking make them unsuitable for a distributed drone system requiring continuous communication.

More powerful platforms such as the Raspberry Pi 4 and Jetson Nano were also considered. These devices offer substantially higher computational performance and support full operating systems; however, they introduce unnecessary overhead for this application. Their increased size, weight, and power consumption negatively impact flight time and payload capacity, which are critical constraints for aerial systems. Additionally, the processing capabilities of these platforms exceed what is required for flight control and communication, making them inefficient choices.

The ESP32-S3 provides the necessary functionality while maintaining a compact form factor and low energy consumption, making it well-suited for deployment across multiple drones. Its cost-effectiveness also supports scalability, allowing the system to expand without significantly increasing overall project cost. For these reasons, the ESP32-S3 was selected as the primary microcontroller for the Aerial Matrix™ system.

3.2.2 AI Acceleration

For the Aerial Matrix™ system, several different AI accelerators were considered based on their white paper YOLO inference performance, efficiency, cost, and ease of use with modern AI standards. Because of the need for modular and flight capable deployment, the accelerator needs to maintain low power consumption, light weight, and reliable communication. Table 3-2 shows the specifications of the processing platforms considered.

Table 3-2: AI Acceleration Options

Processor	Inference Time (ms)	Connectivity	Power Consumption (watts)	Price (USD)	Suitability
Requirements	≤ 100	I^2C or SPI preferred	≤ 5	< $100	Lightweight, ease of use
Grove Vision V2[6]	40	SPI, I^2C, UART	0.35	$30	Excellent
Coral Micro [7]	50	WI-FI, UART	3	$80	Excellent
Arduino Uno Q [8]	100	WI-FI, UART	3	$22	Okay
Raspberry Pi4 Model B 4GB [7]	300-3670	WI-FI, UART	3 – 7	$60 - $110	Poor
Orin Jetson Nano Super [7]	16	WI-FI, UART	5 – 10	$150 - $200	Okay
AWS Cloud [9]	25	Internet Conn.	N/A	$0.526/hr	Okay

The Grove Vision AI v2 Kit was selected as our primary AI Acceleration solution to be used in the Aerial Matrix™ system due to its balance of cost, efficiency, performance, and ease of use. Additionally, it integrates with our previous choice of microcontroller being the ESP32-S3.

Although other options have better performance, they have trade-offs that were unappealing. The Orion Jetson Nano Super is extremely performance, especially for its power draw. The issues with implementing it into our design would require major overhaul in physical constraints. Additionally, the cost was too much to justify for our project. Another performance option was AWS Cloud Object Detection, but it requires continuous internet and a subscription service, which would defeat the idea of an autonomous self-independent drone system.

Other options that were not chosen were the Raspberry Pi 4 Model B and Arduino Uno Q for its lack of object detection inference performance. Both of these were not up to par for our performance requirements. Although the Arduino Uno Q does technically meet the requirements, it is not nearly as performant as the other options.

We are still exploring the Google Coral Dev Micro as an alternative option to the Grove Vision AI v2 kit. This decision will be decided during field testing.

3.3 Subsystems

The design for Aerial Matrix™ is divided into five primary subsystems. These subsystems are responsible for enabling autonomous flight, communication, modular payload integration, and object detection. The subsystems are listed as follows:

1. Drone Construction Subsystem – responsible for the physical structure and flight hardware
2. Drone Control Subsystem – manages stabilization and autonomous positioning
3. Drone Network Subsystem – enables communication between drones
4. Attachment Subsystem – supports modular payload integration
5. Object Detection Subsystem – provides target identification and tracking

3.3.1 Drone Construction

The Drone Construction subsystem allows for the housing and transportation of other subsystems. It also allows for the project's flight capabilities. This subsystem must be durable enough to withstand mild weather conditions, as well as being light enough to allow for the flight of the design. It also encompasses the powering of our drones for adequate use. The options are compared in Table 3-3 as shown below.

Table 3-3: Drone Frame Options

Product	Camera Support	Material	Takeoff Weight (g)	Wheelbase (mm)	Cost (USD)
Requirements	Integration able	Hard Plastic Strength (Minimum)	~1300	~500	< $100
SpeedBee Mario 5 [10]	Yes	Aviation Aluminum	Not found	226-227	$50
F450 Quadcopter [11]	Allows for integration	Plastic	800-1200	400-450	$24
Tarot M690A [12]	Unknown	Aluminum	4320	690	$100-2,000
GEPRC Frame Kit [13]	Yes	Carbon Fiber	Minimal	200-250	$27

The F450 quadcopter frame was selected based on its balance of cost, weight, and structural adequacy for the system's payload requirements. While higher-grade materials such as aluminum and carbon fiber offer increased strength, they introduce unnecessary cost or brittleness for the intended operating conditions. The selected frame provides sufficient structural integrity to support onboard components, including motors, battery, sensors, and modular attachments, while maintaining a lightweight design that supports efficient flight performance. Additionally, its open structure enables straightforward integration of custom attachment mechanisms, aligning with the system's modular design requirements. The F450 Quadrocopter frame is shown in Figure 3-3 below.

Fig 3-4: F450 Quadcopter Frame

3.3.2 Drone Controls

The Drone Control subsystem enables drone control via a standardized HTTP API interface allowing for scalable drone control distribution, low latency control response, and future security considerations via TLS. This HTTP connection will be served over the Drone Network subsystem. This subsystem must provide stable flight, defined as stationary hovering flight with up to ±0.5m deviation from intended location provided for 5 minutes. The Drone Controls facilitates the flight capabilities of the drone and enables intelligent orchestration.

The Drone Control subsystem includes the flight control thread which runs once every 4ms for a stable 250hz inference. The flight control will consist of four modes: *STOP*, *STABILIZE*, *ACRO*, and *LANDING*. The *STOP* flight mode will keep all inputs and motor modes to zero to maintain a safe stationary implementation. This flight mode will be used during transportation, charging, non-flight, and other situations where the drone is not in operational use. The *STABILIZE* flight mode will maintain altitude and position based on the implemented gyroscope and barometer. This flight mode will be used once the drone is in the desired position. The *ACRO* flight mode actively moves the drone based on the control variables provided via drone control commands, gyroscope, and barometer. This is to be used during repositioning. The *LANDING* flight mode is functionally the same as *STABILIZE* but with a de-throttle function which will steadily drop the throttle until it is zero. This is to be used in functionally emergency situations, such as motor failure, extended loss of connection, and extremely low battery levels.

The control system utilizes a closed-loop feedback architecture combining sensor fusion, via Kalman calculation, and proportional–integral–derivative (PID) control. Sensor inputs from the gyroscope and barometer are processed through a filtering mechanism to estimate system state, which is then used by the PID controller to compute corrective motor outputs. This approach enables stable flight, disturbance correction, and precise positional control under varying environmental conditions.

Additionally, a time-series SQL based database will be provided via the Drone Control's HTTP API. This will provide data such as peer-to-peer distances, altitude, performance metrics (motors, controller, etc.), and logging provided via the individual threads.

3.3.3 Drone Network

The Drone Network subsystem enables communication between drones within the Aerial Matrix™ system, allowing for coordinated operation and data exchange. This subsystem must support communication over a minimum distance of 15 meters while maintaining a data transmission latency below 500 ms and a bandwidth of at least 5 Mbps. The network facilitates the transmission of control commands, telemetry data, and target information between the parent drone and child drones. Reliable communication is critical to ensure synchronization and prevent errors in positioning or task execution. A comparison between the protocols are provided in Table 3-4 shown below.

Table 3-4: Communication Protocol Options

Protocol	Range(m)	Latency (ms)	Bandwidth (Mbps)	Complexity
Requirements	≥ 15	≤ 500	≥ 5	Minimize
Wi-Fi	15 - 50	10 - 200	10 – 100+	Medium
ESP-NOW	10 - 30	1 - 50	1 - 10	Low
Bluetooth	5 - 15	20 - 100	0.5 - 3	Low

A Wi-Fi-based communication system was selected for this subsystem due to its ability to meet the required range, bandwidth, and latency specifications. The ESP32-S3's built-in Wi-Fi capabilities allow for direct implementation of wireless communication without additional hardware, reducing system complexity and weight. Alternative communication methods, such as Bluetooth, were considered but rejected due to their limited range and bandwidth. Lower-level communication protocols also lacked the scalability required for multi-drone coordination. The use of IEEE 802.11-based communication ensures reliable and scalable connectivity, making it well-suited for distributed drone systems operating in dynamic environments.

3.3.4 Attachments

The Modular Attachments subsystem enables the Aerial Matrix™ to support interchangeable payloads that expand system functionality without requiring changes to the core drone platform. This subsystem must allow attachments to be installed or removed with one hand within 20 seconds while maintaining secure operation during flight. Attachments must also survive a 2-meter drop test without detaching, ensuring mechanical robustness during field deployment and emergency landings. Reliable attachment performance is essential for enabling rapid mission reconfiguration, modular sensing, and the integration of high-performance computational units such as the Grove Vision v2 or the Google Coral TPU.

Alternative attachment methods, such as magnetic mounts or threaded fasteners, were evaluated but rejected. Magnetic systems lacked the mechanical retention required to survive the 2-meter drop test, while threaded fasteners significantly increased attachment time, violating the system's usability requirements. The selected latch-based design provides a balance of speed, reliability, and mechanical strength, making it well-suited for modular drone systems operating in dynamic and unpredictable environments. Several attachment methods were evaluated, including magnetic mounts, threaded fasteners, and latch-based mechanisms. Each approach was assessed based on installation time, mechanical strength, and reliability under dynamic conditions.

A standardized latch-based mounting mechanism was selected for this subsystem to meet the required interchangeability and durability specifications. The design ensures that attachments remain secured under acceleration, vibration, and minor collisions, while still allowing rapid manual interchangeability in field operations. The mechanism provides a repeatable, tool-free interface that ensures consistent alignment between the drone and the attachment. Electrical connectivity is provided through a regulated 5 V power rail and a digital communication interface, typically UART or SPI, enabling attachments to exchange data with the ESP32-S3 flight controller. This architecture supports a wide range of payloads, including illumination modules, high-performance object detection units, and specialized sensing packages.

The use of a standardized, rugged attachment interface ensures compatibility across multiple payload types, supports rapid field deployment, and aligns with the system requirements for modularity, scalability, and operational efficiency. Figure 3-5 details the part that will be attached to the drone, allowing the attachment to be slid on and off of the drone. It will feature a toggle latch to secure the part.

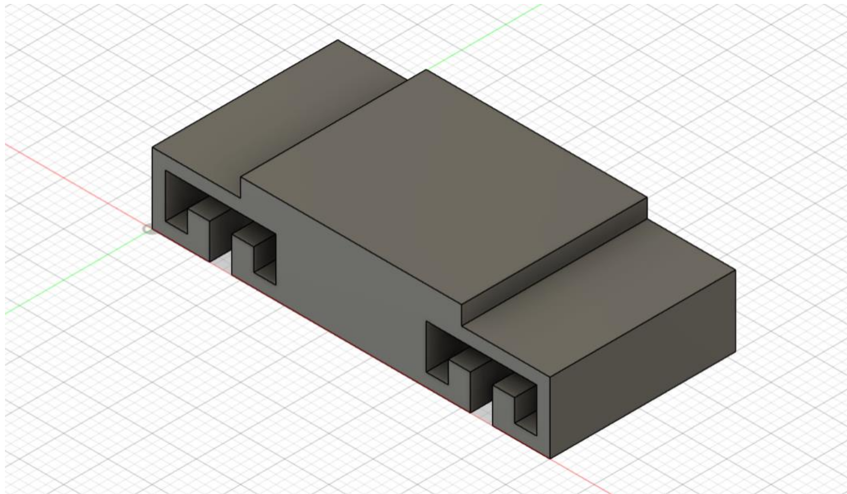

Fig 3-5: Drone-Side of the Attachment System

3.3.5 Object Detection

The Object Detection subsystem provides the Aerial Matrix™ with the capability to identify and track targets within the operational environment. This subsystem must detect at least three objects measuring 30 in × 30 in with a minimum accuracy of 65% and within a five-second processing window. Detection must occur from an overhead perspective at distances ranging from 2 to 10 meters, enabling the system to locate individuals or points of interest during low-visibility or obstructed search conditions. Reliable detection performance is essential for enabling autonomous positioning, coordinated illumination, and mission-critical decision-making across the drone swarm.

Figure 3-6 illustrates the step-by-step process used by the Object Detection subsystem—from camera input through preprocessing, YOLO inference, filtering, and control output—to achieve real-time target identification.

Fig 3-6: Object Detection Process

The object detection process begins with image acquisition from the onboard camera, followed by preprocessing to normalize input data. The processed image is then passed into a YOLO-based neural network model, where inference is performed to identify objects and generate bounding boxes with associated confidence scores. Post-processing filters are applied to remove low-confidence detections, and the resulting target data is transmitted to the control subsystem to inform positioning and tracking behavior. This pipeline enables near real-time detection while maintaining system efficiency. The primary AI acceleration solution is shown in Figure 3-7 below.

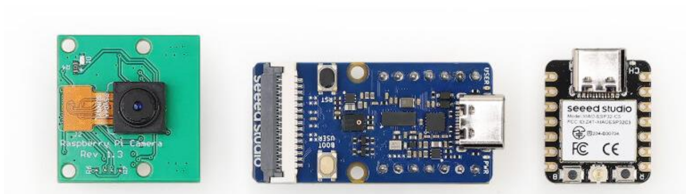

Fig 3-7: Grove Vision AI v2

A modular, high-performance computing approach was selected for this subsystem to meet the required accuracy and response time. Instead of relying on the ESP32-S3 for onboard inference, object detection is executed on a dedicated attachment containing either a Grove Vision v2 or a Google Coral TPU. These platforms provide significantly higher computational throughput, enabling real-time inference using modern convolutional neural networks such as YOLO-based architectures. Offloading detection to a modular attachment reduces the processing burden on the drone, lowers per-drone cost, and supports redundancy by allowing multiple drones to share detection data through the network.

Alternative approaches, such as performing detection directly on the ESP32-S3 or offloading all processing to a ground station, were rejected. The ESP32-S3 lacks the computational capability required for multi-object detection at the specified accuracy and speed, while ground-station processing introduces latency, increases communication overhead, and reduces reliability in remote environments. The Grove Vision AI v2 and Coral TPU provide a balanced solution by offering high-performance inference within a compact, lightweight module that integrates cleanly with the modular control architecture.

The use of a dedicated detection attachment ensures scalable, distributed sensing across the Aerial Matrix™ while maintaining responsiveness, accuracy, and compatibility with the system's modular payload design.

3.4 Level 2 Prototype Design

Our intentions for our completed project in Design II will be to integrate our Object Detection subsystem and our Modular Attachments subsystem. We will also incorporate 3 more drones into the matrix for demonstration of its full capabilities. The attachment mechanism will be custom designed via 3D printing to allow for interchanging of a variety of custom modules.

3.4.1 Level 2 Diagram

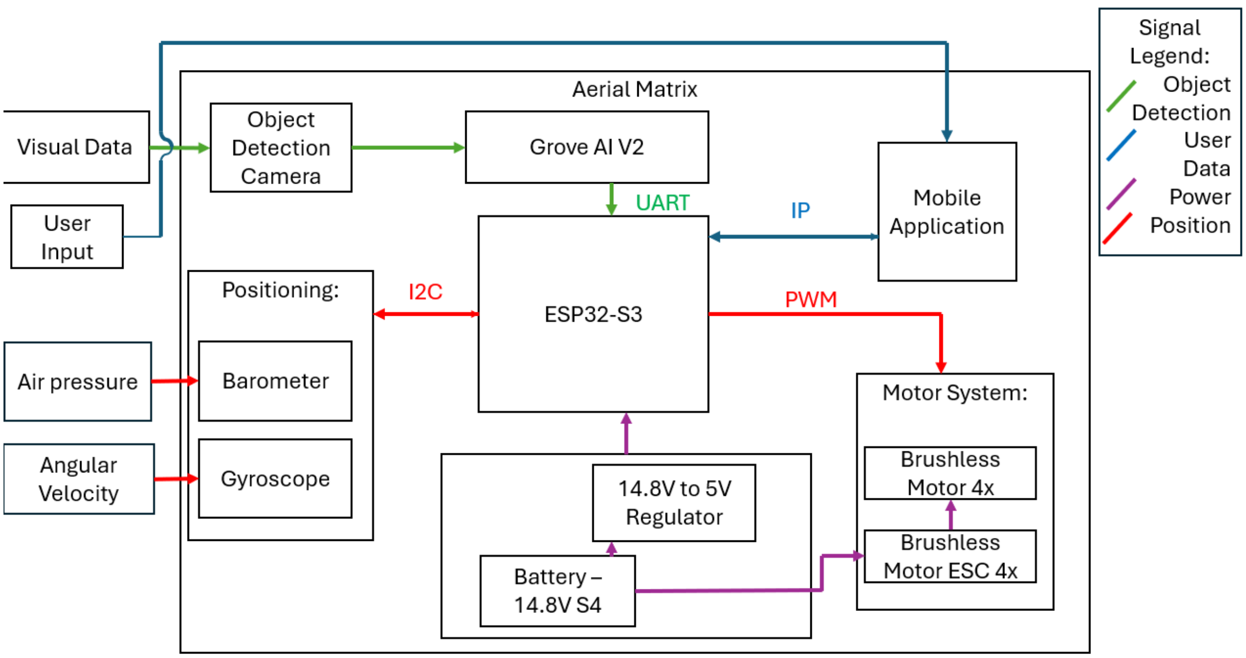

Fig 3-8: Diagram for Aerial Matrix™ (Level 2)

3.5 References

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